

# Scope of Work (SoW) - v2

## Automated Travel Plan Creator Engine Implementation Framework

### 1. Project Overview & Objective

The objective of this project is to build an **Automated Agentic Travel Plan Creator** that translates highly specific user parameters into completely personalized, multi-day vacation packages. Using real-time live search tools, the system aggregates optimal flight connections, designs hyper-personalized day-by-day itineraries (including sight-seeing, shopping hotspots, and curated culinary stops), and maps an accurate aggregate cost estimate.

#### Captured User Input Matrix:

- **Core Parameters:** Destination, Precise Dates, Total Passenger Count (Pax), & Staying Preference (Hotels, BNB, Resorts).
- **Demographic Dynamic:** Travel Group Class (Family Friendly, Couple, or Friends).
- **Dietary & Lifestyle:** Food Preferences (Veg, Non-Veg, Vegan, Jain) & Beverage Preferences (Alcohol / Non-Alcohol).
- **Budget Boundary:** Overall Budget caps (Tiered/Literal Limits).

To evaluate processing flexibility, tool accuracy, and context handling, the project will implement two architectural tracks:

1. **Approach 1 (n8n Workflow):** An enterprise low-code integration system tracking state via visual nodes.
2. **Approach 2 (LangGraph & LangChain Blueprint):** A highly granular code-defined state machine managing deterministic fallback reasoning loops.

### 2. Approach 1: Low-Code Automation Workflow Engine via n8n

This path maps out a scalable, visually componentized travel coordinator engine on the n8n canvas using standard integrations and advanced structural AI sub-nodes.

#### Architecture & Component Mapping

- **Input Layer:** An active Chat Trigger or Webhook Node parses the user matrix (Destination, Dates, Dietary rules, Budget limits, etc.).
- **Orchestrating Brain:** An AI Agent Node backed by an LLM (such as GPT-4o mini) dynamically determining which micro-tool needs execution based on prompt context.

- **Context Window Memory Node:** Holds structural dialogue traces across chat interactions to seamlessly modify itineraries based on live adjustments (e.g., "Swap Day 2 with more shopping").
- **Integrated Tooling Suite:**
  - **SerpAPI / Google Search Block:** Dispatched dynamically to scrape real-time regional flight offers matching exact dates, spot regional shopping malls/markets, and verify food options adhering strictly to the user's dietary tags (e.g., searching for "Jain food options" or "Vegan friendly" restaurants in the destination).
  - **Calculator Node:** Aggregates live numbers scraped from flight data, estimated meal tariffs, entry tickets for sightseeing, and lodging to produce a clear Estimated Budget breakdown.

## Key Deliverables

- Fully packaged n8n canvas workflow configuration schema.
- Secure authorization mappings for operational keys (SerpAPI & OpenAI).
- Functional web chat widget for user sandbox testing.

## 3. Approach 2: Granular Custom Engine via LangGraph & LangChain

This developer-first pipeline structures a cyclic state-machine application, giving absolute programmatic control over complex dependency parsing (e.g., ensuring flight dates match lodging availability).

### State Engineering & Node Workflows

- **Central System State (TravelState Schema):** A structured schema capturing raw inputs, verified flight listings, day-by-day structured nodes, dietary constraints, and rolling financial ledgers.
- **Custom Node Architecture:**
  - **Intent & Parser Node:** Structurally maps inputs into distinct semantic variables, verifying demographic suitability (e.g., flagging bars if "Family Friendly" is requested alongside "Alcohol").
  - **Flight Sourcing Tool Node:** Uses LangChain community wrappers over search engines to grab actual live flight costs and route choices for the selected dates.
  - **Itinerary Synthesizer Node:** Sequentially queries and populates each day's layout with Sightseeing Locations, Shopping Outlets, and specialized Food Options.
  - **Financial Validator Node:** Compiles cost matrices. If the total breaches the user's budget ceiling, a conditional edge reroutes execution back to the synthesizer node with constraints to pick more economical choices.

## Key Deliverables

- Python codebase managing the underlying LangGraph state network.
- Container-ready environment configurations (Dockerfile / FastAPI setup).
- Interactive script suite to generate structured JSON or Markdown itineraries.

## 4. Feature Matrix Comparison

| Feature Element             | Approach 1: n8n Engine                                          | Approach 2: LangGraph Stack                                            |
|-----------------------------|-----------------------------------------------------------------|------------------------------------------------------------------------|
| Input Validation            | Handled via prompt parsing instructions inside the agent block. | Enforced programmatically via Pydantic model state boundaries.         |
| Budget Overrun Optimization | Approximated via prompt adjustments in successive loops.        | Deterministic cyclic loops that re-execute nodes until criteria match. |
| Execution Traceability      | Visual tracking showing precisely which tool node fired.        | Granular step logging using LangSmith telemetry.                       |

## 5. Project Timeline

- **Week 1: Schema Design & Setup:** Mapping user schemas, structuring prompt guardrails, and setting up API permissions.
- **Week 2: n8n Flow Assembly:** Wiring search tools, embedding custom filters for dietary preferences, and validating memory limits.
- **Week 3: LangGraph Coding & Optimization:** Implementing cyclic state nodes, conditional budget boundaries, and flight aggregations.
- **Week 4: Review & Deploy:** Stress-testing diverse user matrices, tuning itinerary formatting, and delivery sign-off.

## 6. References

- **n8n AI Agents & Workflow Archetype:** [AI Learners India System Setup](#)
- **n8n Developer Platform:** [n8n Official Website](#)
- **Cyclic State Orchestration:** [LangGraph Framework Manual](#)